UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 07811

LH GENERAL PHYSICS

SUMMER EXAMINATION 2024

Time Allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

All symbols have their usual meanings.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. An electron is moving in a one-dimensional infinite square well of width a as described by the time-independent Schrödinger equation:

$$\left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x) = E\psi(x),$$

where:

$$V(x) = \begin{cases} 0 & -a/2 \leq x \leq a/2\ , \\ \infty & \text{otherwise,} \end{cases}$$

and all other symbols have their usual meaning.

(a) Without explicit calculation, sketch the potential and the form of the corresponding lowest three energy eigenstates. **[5]**

(b) Consider the two wavefunctions:

$$\psi_\alpha = \begin{cases} A\cos(5\pi x/a) & -a/2 \leq x \leq a/2\ , \\ 0 & \text{otherwise,} \end{cases}$$

$$\psi_\beta = \begin{cases} B\sin(26\pi x/a) & -a/2 \leq x \leq a/2\ , \\ 0 & \text{otherwise,} \end{cases}$$

where A and B are normalization constants. Show that both ψ_α and ψ_β are eigenstates of the above Schrödinger equation and that they satisfy the appropriate boundary conditions. What are their associated energies? **[8]**

(c) Determine the value of A for the wavefunction ψ_α in part (b). **[5]**

(d) When an electron in state ψ_β transitions into state ψ_α, the excess energy is released in the form of a photon. If the width of the potential well is given by $a = 10.0$nm, calculate the wavelength of this emitted photon. **[5]**

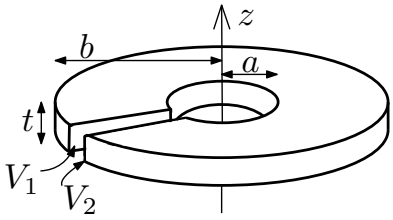
(e) If an electron moving in free space has the same energy as the photon in part (d), what would be its corresponding de Broglie wavelength? **[3]**

(f) From your results in part (d) and (e), would you expect an electron microscope or an optical microscope at this energy to have higher resolution? **[1]**

(g) What would the de Broglie wavelength be if the electron moving in free space instead had an energy of 1000keV? **[3]**

2. (a) Show that the magnetic field B at the centre of a circular current loop of radius r and current I is given by $B = \mu_0 I/2r$. Use a sketch to show the direction of the field. **[5]**

(b) An annular copper plate of resistivity ρ is used as part of an electromagnet. It has an inner radius a, outer radius b, thickness t and has a narrow radial slot cut through it. The faces of the slot are held at potentials V_1 and V_2, causing a current to circulate.



By considering its subdivision into a nest of rings, or otherwise, obtain an expression for the magnetic field produced at the centre of the annulus. **[7]**

(c) Sketch a graph to show approximately how the magnetic field varies with position along the z axis of the annulus in (b). The field is to be used to lift a small permanent magnet of mass m and magnetic moment μ against the force of gravity in the $-z$ direction. The magnet remains on the z axis with its moment aligned with the field. Sketch how the magnetic force varies with position, and indicate where the gravitational and magnetic forces on the permanent magnet will be balanced. **[5]**

(d) Suppose a long, solenoid-like electromagnet is constructed by stacking a large number of the plates in part (b) on their common z axis.
(i) Explain what electromagnetic forces are present within and between the plates when the current flows.
(ii) A larger field requires a larger current. Explain how the electrical power dissipated in the electromagnet scales with the magnitude of the field produced, and why most dissipation will occur near its inner radius. **[6]**

(e) Thermally, the electromagnet in part (d) can be considered as a a copper tube of length l, wall thickness $(b - a)$ and thermal conductivity κ. Obtain an expression for the thermal resistance between the inside and outside of the tube. Comment on the merits of cooling the outside of the electromagnet by (i) exposure to the air, (ii) a stream of water, and (iii) thermal radiation. **[7]**

3. (a) Consider a planar surface that is defined by $z = 0$ in a cartesian space, with the force of gravity acting in the negative z-direction. A particle of mass m moves on this surface in the positive x-direction towards the origin, at a constant speed v. When the particle arrives at $x = a$ it encounters a step such that the surface abruptly drops to $z = -b$ for $x > a$.

   i. Draw a labelled sketch of the x-z plane of this system, showing the surface and the motion of the particle. **[3]**

   ii. Show that the particle lands on the surface at a location given by the following expression:

$$(x, y, z) = \left(a + v\sqrt{\frac{2b}{g}}, 0, -b\right),$$

   where g is the gravitational acceleration. **[3]**

(b) Consider the same particle travelling on the same horizontal surface. Now, when the particle reaches $x = a$, the surface changes abruptly to be described by the following function:

$$z = b\left(\frac{x-c}{a-c}\right)^2 - b,$$

where $c > a$.

   i. Draw a labelled sketch of the x-z plane of this system, showing the the surface and the motion of the particle. **[4]**

   ii. Show that the x-coordinate of the particle when it strikes the surface is given by:

$$x = a + \frac{4bv^2(c-a)}{g(c-a)^2 + 2bv^2}$$

[8]

(c) Consider again the same particle travelling on the same horizontal surface. Now, at $x = a$, the surface changes smoothly and continuously to be described by $z = -d(x-a)^2$, where d is a constant with units of m^{-1}. Derive and fully justify the condition that must be satisfied for the particle to leave the surface at $x > a$. **[7]**

(d) State any assumptions that you have made in answering parts (a)-(c). **[5]**

Formula Sheet

Vector notation is shown only in essential cases.

$$F = ma = \frac{\mathrm{d}p}{\mathrm{d}t}$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = \boldsymbol{r} \times (m\boldsymbol{v})$$

$$\boldsymbol{\tau} = \boldsymbol{r} \times \boldsymbol{F}$$

$$\tau = I\frac{\mathrm{d}\omega}{\mathrm{d}t}$$

$$I = mr^2$$

$$I_P = I_{cm} + Mr^2$$

$$F_c = mv^2/r$$

$$a_r = \frac{v^2}{r} = \omega^2 r$$

$$\boldsymbol{F} = -\boldsymbol{\nabla}U \quad \text{or} \quad -\frac{\mathrm{d}U}{\mathrm{d}x}$$

$$K = \frac{1}{2}I\omega^2$$

$$\omega_0 = 2\pi f_0 = \sqrt{\frac{k_s}{m}}$$

$$F_G = \frac{Gm_1m_2}{r^2}$$

$$U = mgh$$

$$\boldsymbol{E} = \frac{Q}{4\pi\epsilon_0 r^2} = -\boldsymbol{\nabla}V$$

$$V = -\frac{Q}{4\pi\epsilon_0 r}$$

$$\mathrm{d}\boldsymbol{B} = \frac{\mu_0 I\,\mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}}}{4\pi r^2}$$

$$\boldsymbol{\nabla} \times \boldsymbol{B} = \mu_0\left(\boldsymbol{J} + \frac{\partial \boldsymbol{D}}{\partial t}\right)$$

$$\boldsymbol{\nabla} \times \boldsymbol{E} = -\frac{\partial \boldsymbol{B}}{\partial t}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{E} = \frac{\rho}{\epsilon_0}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{B} = 0$$

$$\Phi = \int_S \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{A}$$

$$\epsilon = -\frac{\partial \Phi}{\partial t}$$

$$\oint \boldsymbol{E} \cdot \mathrm{d}\boldsymbol{A} = \frac{Q}{\epsilon_0}$$

$$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 \boldsymbol{I}$$

$$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$$

$$\boldsymbol{B} = \mu_0(\boldsymbol{H} + \boldsymbol{M})$$

$$\boldsymbol{D} = \epsilon_0\boldsymbol{E} + \boldsymbol{P}$$

$$u = \frac{1}{2}\epsilon_0 E^2$$

$$u = \frac{1}{2}\frac{B^2}{\mu_0}$$

$$R = \rho L/A$$

$$P = VI = I^2R = V^2/R$$

$$C = Q/V; \quad C = \epsilon_0 a/d$$

$$\Phi = LI$$

$$\Phi_2 = M_{21}I_1$$

$$U = \frac{Q^2}{2C}$$

$$U = \frac{1}{2}LI^2$$

$$PV = nRT$$

$$C = \frac{\mathrm{d}Q}{\mathrm{d}T}$$

$$\gamma = \frac{C_P}{C_V}$$

$$PV^\gamma = K$$

$$v = \sqrt{\frac{\gamma RT}{m}}$$

$$\mathrm{d}U = đW + đQ$$

$$S = k_B \ln \Omega$$

$$\mathrm{d}S \geq \frac{đQ}{T}$$

$$P(E_n) = \frac{g_n}{Z} \exp\left(-\frac{E_n}{k_B T}\right)$$

$$\dot{Q} = kA\boldsymbol{\nabla} T \quad \text{or} \quad kA\frac{\mathrm{d}T}{\mathrm{d}x}$$

$$\dot{Q} = \epsilon\sigma A T^4$$

$$\lambda_{\max} T = K$$

$$v_p = f\lambda = \frac{\omega}{k}$$

$$v_g = \frac{\partial\omega}{\partial k}$$

$$k = 2\pi/\lambda$$

$$\omega = 2\pi f$$

$$Y = Y_0 \sin(kx - \omega t)$$

$$Y = Y_0 \exp i(kx - \omega t)$$

$$v = \sqrt{Y/\rho} \quad \text{or} \quad \sqrt{T/\mu}$$

$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \frac{v \pm v_{\text{rec}}}{v \pm v_{\text{src}}}$$

$$2d \sin\theta = n\lambda$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}}$$

$$x' = \gamma(x - ut)$$

$$t' = \gamma(t - ux/c^2)$$

$$v' = \frac{v-u}{1-uv/c^2}$$

$$p = \gamma m_0 v$$

$$E^2 = p^2c^2 + m_0^2c^4$$

$$\frac{f_{\text{rec}}}{f_{\text{src}}} = \sqrt{\frac{1-\beta}{1+\beta}}$$

$$E = hf$$

$$p = \frac{h}{\lambda}$$

$$N = N_0 \exp(-t/\tau) = N_0 \exp(-\lambda t)$$

$$\Delta p \cdot \Delta x \geq \hbar$$

$$\Delta E \cdot \Delta t \geq \hbar$$

$$\hat{\boldsymbol{p}} = -i\hbar\boldsymbol{\nabla}$$

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

$$c = 1/\sqrt{\mu_0\epsilon_0}$$

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

$$n = c/v$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\sin\theta = 1.22\,\lambda/d$$

$$d\sin\theta = m\lambda$$

$$l = l_0(1 + \alpha\,\Delta T)$$

$$E = \frac{F/a}{\Delta l/l}$$

$$\gamma = F/d = W/a$$

$$p = \rho g h$$

$$F_{\text{up}} = (\rho_2 - \rho_1) g v$$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^3 W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^8 m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

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Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.